

Don't Think of the White Bear: Ironic Negation in Transformer Models Under Cognitive Load

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Abstract

Negation instructions such as 'do not mention X ' can paradoxically increase the accessibility of X in human thought, a phenomenon known as ironic rebound. Large language models (LLMs) face the same challenge: suppressing a concept requires internally activating it, which may prime rebound instead of avoidance. We investigated this tension with two experiments. **(1) Load & content:** after a negation instruction, we vary distractor text (semantic, syntactic, repetition) and measure rebound strength. **(2) Polarity separation:** We test whether models distinguish neutral from negative framings of the same concept and whether this separation predicts rebound persistence. Results show that rebound consistently arises immediately after negation and intensifies with longer or semantic distractors, while repetition supports suppression. Stronger polarity separation correlates with more persistent rebound. Together, these findings, complemented by a circuit tracing analysis that identifies sparse middle-layer attention heads amplifying forbidden tokens while early layers suppress, link cognitive predictions of ironic rebound with mechanistic insights into long-context interference. To support future work, we release ReboundBench, a dataset of 5,000 systematically varied negation prompts designed to probe rebound in LLMs.

1 Introduction

Human attempts to suppress thoughts can paradoxically increase their accessibility, a phenomenon known as **ironic rebound** [22, 21, 8]. Classic demonstrations (e.g., "do not think of a white bear") show negation may highlight the forbidden concept [23]. Cognitive accounts attribute this effect to semantic priming and psychological reactance: monitoring processes bring the target concept to mind to check for compliance, and restrictions on thought can provoke a counteractive focus [16, 13, 18, 15]. These findings suggest a general principle: suppression under load can backfire [22].

Large language models (LLMs) face a similar interpretive challenge. To follow an instruction like "do not mention X ," they activate an internal representation of X . Intuitively, this activation could produce rebound-like behavior, making the forbidden concept more likely rather than less [2]. Prior studies of negation in LLMs support this intuition, showing that models occasionally fail to respect negated instructions [20, 19]. Such failures are not only theoretically interesting but also practically significant for safety and moderation, where filtering rules rely on negation.

At the same time, mechanistic interpretability work points to architectural forces that may resist rebound. Certain attention heads (e.g., L10H7 in GPT-2 Small) implement a copy suppression mechanism that reduces the probability of tokens that are deemed forbidden (e.g., due to negation instructions)[11]. Prior work further shows that LLMs often struggle to retrieve information placed in

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the middle of long contexts - a phenomenon known as lost-in-the-middle [10]. These counterforces suggest that, unlike humans, LLMs may display systematic suppression effects [12]. This tension raises the question: under what conditions do models show rebound, suppression, or both?

Existing evaluations leave this question unresolved. Prior work often blurs lexical repetition avoidance with genuine semantic negation, and rarely manipulates distractor text beyond trivial additions of text [10, 7]. We lack clarity on how intervening load affects negation in LLMs. This gap is consequential: if suppression is fragile to paraphrase or distraction, moderation systems based on simple negation heuristics may be unreliable.

In this paper, we make two main contributions. First, we introduce ReboundBench, a dataset of 5,000 negation prompts with systematically varied distractor types (semantic, syntactic, repetition) and lengths, enabling controlled study of ironic rebound dynamics in LLMs. Second, through two experiments (i) varying distractor load to test when suppression weakens and rebound emerges, and (ii) measuring polarity separation across framings. We show how rebound strength and persistence vary across model families and scales and verify this through circuit tracing analysis. We evaluate nine open-source models ranging from GPT-2 Small to GPT-OSS-20B [14, 4, 24, 26, 6, 25, 5, 1, 9].

2 Methods

2.1 Preliminaries

Log-probability Each model’s native tokenizer is used; when a candidate target concept X tokenizes into multiple subwords, the item is either replaced with a single-token synonym or excluded, depending on availability. We evaluate, $\log p_\theta(X \mid c) = \log \prod_{t=1}^T p_\theta(x_t \mid c, x_{<t})$, where $X = (x_1, \dots, x_T)$ is the tokenized form of the target concept. Since our dataset restricts X to single tokens, this simplifies to a single conditional log-probability. Outliers and skipped items (multi-token X without synonym replacement) are excluded consistently across conditions.

Load We implemented load (ℓ) as the amount of distracting texts that the model must process. To account for different context windows we converted the raw lengths of the distractors to within-model percentiles; $\ell = \text{rank_pct}(\text{distractor_length})_{\text{within model}} \in [0, 1]$ where $\ell = 0$ represents minimal text between instruction and measurement, and $\ell = 1$ represents maximum text. To properly compare the effects in log-probability described below, we establish a **high-load baseline**. By calculating the mean log-probability when the cognitive load is the highest (top 5% of ℓ), denoted by $\mu_{\text{base}} = \mathbb{E}[\log_2 p(\text{forbidden}) \mid \ell \geq 0.95]$. At these points, the negation instruction is maximally attenuated, revealing the model’s natural probability for the forbidden token.

Surprisal To quantify the ironic rebound effect, we measure the change in the surprisal of the forbidden token relative to a high-load baseline. Surprisal, s , measures unexpectedness in bits: $s(x|c) = -\log_2 p_\theta(x|c)$. We define the surprisal difference, $\Delta s(\ell)$, as the change in surprisal at a given load percentile compared to the baseline surprisal under maximum load, $s_{\text{base}} = \mathbb{E}[-\log_2 p(\text{forbidden}) \mid \ell \geq 0.95]$. This difference can be expressed in terms of log-probabilities:

$$\Delta s(\ell) = s_{\text{base}} - s(\text{forbidden} \mid \ell) = \log_2 p(\text{forbidden} \mid \ell) - \mu_{\text{base}}$$

Here, μ_{base} is the mean base-2 log-probability at maximum load ($\ell \geq 0.95$). Positive values of $\Delta s(\ell)$ (in bits) indicate ironic rebound—the negation instruction has made the forbidden token less surprising (more probable) than it would be at baseline. Negative values indicate suppression.

Suppression score We quantify suppression versus rebound with a score that compares the log-probability of the target under the suppression and neutral conditions, such that, $S(L) = \mathbb{E}_{(topic, X)} [\log p_\theta(X \mid c_{\text{neu}}(L)) - \log p_\theta(X \mid c_{\text{sup}}(L))]$. Here $c_{\text{neu}}(L)$ is the neutral context with load length L , and $c_{\text{sup}}(L)$ is the suppression context of the same length. $S(L) < 0$ indicates rebound, where the negation instruction increases the likelihood of X .

Polarity discrimination To test whether models distinguish between attitudinal framings, we generate two minimal variants of each context: neutral and negative continuations about the same

concept X and summarize their separation with the following metric:

$$\Delta = \log p_{\theta}(r_{\text{neu}}) - \log p_{\theta}(r_{\text{neg}}). \quad (1)$$

Larger values of Δ reflect stronger polarity discrimination, while smaller values indicate that the model assigns similar probabilities regardless of framing.

Statistical model At the item level, we fit a mixed-effects regression of surprisal difference in bits:

$$\Delta(\text{bits}) \sim \ell + \text{type} + \ell : \text{type} + \text{model} + (1 \mid \text{forbidden_concept}).$$

This specification estimates how rebound strength changes with load, whether collapse dynamics differ by distractor type, and whether effects vary by model family. The random intercept accounts for dependence among items sharing the same forbidden concept. As a robustness check, we also fit cluster-robust OLS regressions. Coefficients of interest are β_{load} (attenuation with load) and $\beta_{\ell \times \text{type}}$ (interaction between load and distractor type).

AUC $_{\Delta}$ AUC is the total rebound mass, integrating positive Δ across the load range

$$\text{AUC}_{\Delta} = \int_0^1 \max(0, \Delta(\ell)) d\ell.$$

2.2 Experimental setup

Models We evaluated nine open-source language models spanning different parameter scales and training regimes: GPT-OSS-20B [1], GPT-Neox-20B [5], Qwen3-14B [25], Llama-3-8B-Instruct [6], OPT-2.7B [26], Bloom-560M [24], Pythia-410M [4], LFM2-350M [9], and GPT-2 Small [14]. This selection covers both base and instruction-tuned models, providing a range of capacities for handling negation and semantic distinctions. Models are accessed via the Hugging Face Transformers library with code publicly available at: <https://github.com/cesium132dot9/Dont-Think-of-the-White-Bear>

Dataset We introduce **ReboundBench**, 5,000 programmatic, templated prompts, publicly available at [Link to Dataset](#). Each prompt pairs a topic with a target concept X (topic supplies local discourse context; X is the negated word). The corpus is balanced across conditions and minimizes hand-selection. All text is lowercased, whitespace-normalized, and punctuation-stripped to reduce tokenization artifacts. Target concepts are common nouns and single tokens under the evaluated tokenizers. The same dataset is used for all models and experiments to enable direct comparison. More details about the dataset are in section A.1.

Evaluation procedure Two experiments are implemented via a custom evaluation script.

- **Suppression under load** Each trial begins with an introduction of the topic and a negation instruction of the form “do not mention X .” An intervening load is then inserted before the evaluation point. Loads vary in both length ($\ell \in \{0, \dots, 1024\}$ tokens, sampled in powers of two) and genre: syntactic fillers (grammatical but semantically light continuations), semantic content (coherent text related to the topic but excluding X), or repetition (restatements of surrounding content without directly naming X). A matched neutral condition omits the negation instruction. We then use the Suppression Score ($S(L)$) and the Surprisal Difference ($\Delta s(\ell)$) metrics to quantify the results.
- **Polarity discrimination** For each target concept, we constructed two context variants (negative and neutral) and measured the model’s ability to distinguish between them using the Polarity Discrimination (Δ) metric. A high Δ indicates that the model assigns clearly different probabilities to the three framings, while a low Δ reflects little separation. As shown in the figures, we plot the distribution of Δ scores across topic–forbidden pairs for several models to compare their capacity for polarity discrimination.

3 Results

Suppression & rebound under load: LLMs mimic ironic rebound when told not to produce a concept, as shown in Figure 1. Semantic distractors trigger the strongest rebound, syntactic

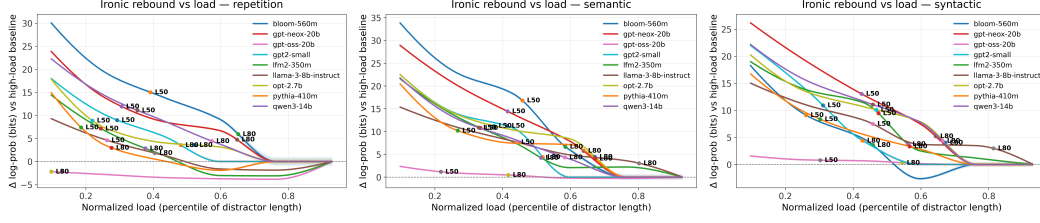


Figure 1: Change in surprisal (Δ bits) of the forbidden token relative to a high-load baseline as normalized distractor length increases: semantic, syntactic, repetition

distractors are weaker, and repetition is the weakest. Smaller models show only brief rebound before suppression, while mid-scale models sustain stronger effects: Bloom remains robust across distractors, Qwen separates semantic from syntactic rebound, and OPT-2.7B combines higher magnitude with persistence. GPT-OSS-20B is an outlier, showing minimal or even negative rebound, breaking the scaling trend. Granular model-level results are provided in the Appendix (Figure B.2 and Table 3).

Polarity discrimination Polarity discrimination is not monotonic with scale. Family effects dominate: Llama-3-8B-Instruct (11.3 bits) and Bloom-560M (10.1) show the strongest separation, while larger models such as GPT-NeoX-20B (5.7) and Qwen-14B (6.5) are weaker. GPT-OSS-20B and Pythia-410M are outliers, the weakest by far. Polarity separation also correlates with rebound persistence L_{50} ($r \approx 0.44$), linking sharper attitudinal encoding to longer-lived rebound. This suggests that models with more nuanced semantic representations may struggle more to suppress activated concepts over long contexts, pointing to a potential tension between representational quality and the stability of cognitive control. Figure 2 shows a clear trade-off where models with a larger rebound magnitude have a longer-lived rebound. Higher AUC tends to pair with higher L_{50} , so models closer to the lower-left are preferable for negation control.

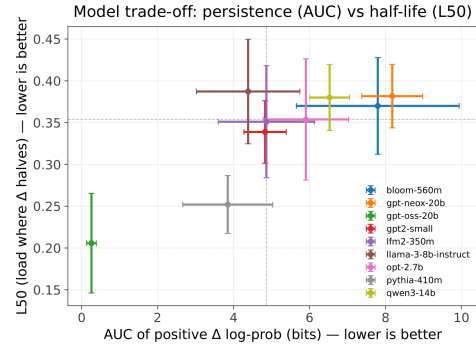


Figure 2: Trade-off between rebound magnitude (AUC) and persistence (L_{50} , the load at which rebound halves)

Internal Mechanisms of Ironic Rebound Having established that language models exhibit ironic rebound when told not to mention specific concepts, we next sought to understand what happens inside the models to produce this effect. We focused on attention heads, specialized components within transformers that determine what information to focus on at each processing step. By systematically disabling individual heads and measuring how this changed the model’s tendency to produce forbidden words, we could identify which heads were most responsible for ironic rebound.

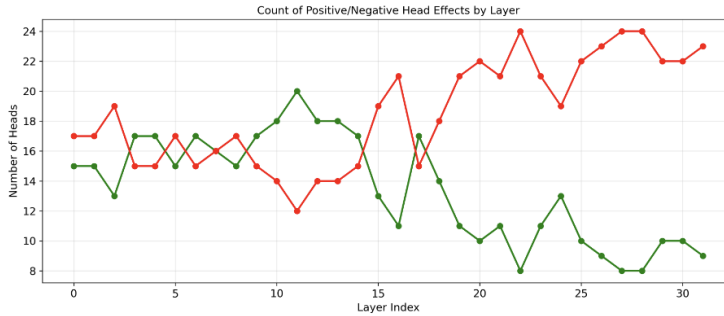


Figure 3: How suppression and amplification effects change across model layers. Early layers suppress, middle layers show mixed effects with emerging rebound, and later layers stabilize.